Topic: Eigenvalues and Eigenvectors

Question 1. Let $\boldsymbol{A} = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

(a) Find the eigenvalues and eigenvectors of $\boldsymbol{A}$.

(b) Verify that the trace equals the sum of the eigenvalues, and the determinant equals their product.

(c) Consider $\boldsymbol{B} = \boldsymbol{A} + 100\boldsymbol{I} = \begin{bmatrix} 101 & -1 \\ 2 & 104 \end{bmatrix}$. Find the eigenvalues and eigenvectors of $\boldsymbol{B}$.

Question 2. Find the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of

$$\boldsymbol{A} = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } \boldsymbol{B} = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}.$$

Check that $\lambda_1 + \lambda_2 + \lambda_3$ equals the trace and $\lambda_1\lambda_2\lambda_3$ equals the determinant.

Question 3. (omitted from AY2023/2024)
Suppose $\mathcal{V}$ is the real vector space of all real-valued functions $f : \mathbb{R} \to \mathbb{R}$ which have derivatives of all orders. Let $\boldsymbol{T} : \mathcal{V} \to \mathcal{V}$ be the linear transformation $\boldsymbol{T}(f) = f''$.
Determine whether each of the following functions f is an eigenvector of $\boldsymbol{T}$? If it is, what is its corresponding eigenvalue?

(a) The constant function $f(x) = k$, where k is a real nonzero constant.

(b) The sine function $f(x) = \sin(kx)$, where k is a real nonzero constant.

(c) The exponential function $f(x) = e^{kx}$, where k is a real constant.

(d) The exponential function $f(x) = \sqrt[3]{x}$.

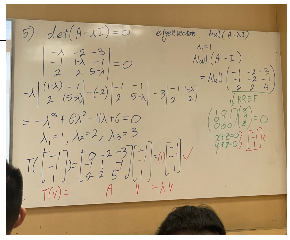
Question 4. Consider the following basis $\mathcal{B}$ for $\mathbb{R}^3$ and the linear operator $\boldsymbol{T}$ on the vector space $\mathbb{R}^3$, where

$$\mathcal{B} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\} \text{ and } \boldsymbol{T}\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = \begin{bmatrix} 3a + 2b - 2c \\ -4a - 3b + 2c \\ -c \end{bmatrix}.$$

(a) Verify that every basis vector in $\mathcal{B}$ is an eigenvector for $\boldsymbol{T}$, and determine its corresponding eigenvalue λ. Here, $\boldsymbol{v}$ is an eigenvector for operator $\boldsymbol{T}$ if $\boldsymbol{T}(\boldsymbol{v}) = \lambda\boldsymbol{v}$ for some value λ.

(b) What is the matrix representation $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}}$ of $\boldsymbol{T}$ with respect to the basis $\mathcal{B}$?

Question 5 continued on next page...



Question 5. Given a matrix $\boldsymbol{A} \in \mathcal{M}_{3\times 3}(\mathbb{R})$, where $\boldsymbol{A} = \begin{bmatrix} 0 & -2 & -3 \\ -1 & 1 & -1 \\ 2 & 2 & 5 \end{bmatrix}$.

(a) Write down the characteristic polynomial of $\boldsymbol{A}$ and determine all the eigenvalues of $\boldsymbol{A}$. You may need to recall the cofactor expansion for determinants.

(b) For each eigenvalue λ of $\boldsymbol{A}$, find the set of eigenvectors corresponding to λ.

(c) Find a basis $\mathcal{B}$ for $\mathbb{R}^3$ consisting of eigenvectors of $\boldsymbol{A}$. Hence, determine an invertible matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$. In this case, we say that $\boldsymbol{A}$ is *diagonalizable*.

Question 6. Consider the linear operator $\boldsymbol{T}$ on the vector space $\mathcal{P}_1(\mathbb{R})$ defined by

$$\boldsymbol{T}(a + bx) = (6a - 6b) + (12a - 11b)x.$$

(a) Obtain the matrix representation $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}}$ of $\boldsymbol{T}$ with respect to the standard basis $\mathcal{B} = \{1, x\}$.

(b) Find all eigenvalues and their corresponding eigenvectors of $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}}$.

(c) Let $\boldsymbol{e}_1 = (a_1, b_1)^T$ and $\boldsymbol{e}_2 = (a_2, b_2)^T$ be two linearly eigenvectors found in (b). Set $\mathcal{A}$ to be the basis $\mathcal{A} = [a_1 + b_1 x, a_2 + b_2]$. What is the corresponding matrix representation $[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}}$?

Question 7. (omitted from AY2023/2024)

Given a matrix $\boldsymbol{A} \in \mathcal{M}_{2\times 2}(\mathbb{C})$, where $\boldsymbol{A} = \begin{bmatrix} i & 1 \\ 2 & -i \end{bmatrix}$.

(a) Determine all the eigenvalues of $\boldsymbol{A}$.

(b) For each eigenvalue λ of A, find the set of eigenvectors corresponding to λ.

(c) Find a basis $\mathcal{B}$ for $\mathbb{C}^2$ (as a complex vector space) consisting of eigenvectors of $\boldsymbol{A}$. Hence, determine an invertible matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$.

Question 8.

(a) Let $\boldsymbol{A}$ be a square matrix. Prove that $\boldsymbol{A}$ and its transpose $\boldsymbol{A}^T$ have the same eigenvalues.

(b) Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be two similar square matrices. Here, $\boldsymbol{A}$ and $\boldsymbol{B}$ are *similar* means there is a matrix $\boldsymbol{P}$ of the same size as $\boldsymbol{A}$ and $\boldsymbol{B}$ such that $\boldsymbol{B} = \boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P}$. Prove that $\boldsymbol{A}$ and $\boldsymbol{B}$ have the same eigenvalues.

Question 9.

(a) A square matrix $\boldsymbol{A}$ is invertible if and only if $\lambda = 0$ is not an eigenvalue of $\boldsymbol{A}$.

(b) Suppose $\boldsymbol{A}$ is an invertible matrix. If λ is an eigenvalue of $\boldsymbol{A}$ and $\boldsymbol{v}$ is a corresponding eigenvector, then $\lambda^{-1} = \frac{1}{\lambda}$ is an eigenvalue of the inverse A^{-1}, and $\boldsymbol{v}$ is a corresponding eigenvector.

Question 10 continued on next page...

John wanted to go thru so i think it's impt

Question 10. Suppose that $\boldsymbol{A}$ is an $n \times n$-matrix with has exactly n eigenvalues (not necessarily distinct), say $\lambda_1, \lambda_2, \ldots, \lambda_n$. In this question, for $n \in \{2, 3\}$, we show that

$$\operatorname{tr}(\boldsymbol{A}) = \sum_{i=1}^{n} \lambda_i \text{ and } \det(\boldsymbol{A}) = \prod_{i=1}^{n} \lambda_i .$$

Recall: $\operatorname{tr}(\boldsymbol{A})$ denotes the trace of $\boldsymbol{A}$. You can follow the general strategy to prove for all n.

(a) Let $n = 2$ and $\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

(i) Show that the characteristic polynomial is of the form $(ad - bc) - (a + d)\lambda + \lambda^2$.

(ii) On the other hand, since $\boldsymbol{A}$ has eigenvalues λ_1 and λ_2, its characteristic polynomial can be written as $(\lambda_1 - \lambda)(\lambda_2 - \lambda)$. Compare with the expression in (a)(i) to show that $a + d = \lambda_1 + \lambda_2$ and $ad - bc = \lambda_1\lambda_2$.

(b) Let $n = 3$ and $\boldsymbol{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$.

(i) Show that the coefficient of λ^2 of characteristic polynomial is $a + e + i$, while the constant term of characteristic polynomial is $\det(\boldsymbol{A})$.

(ii) On the other hand, since $\boldsymbol{A}$ has eigenvalues λ_1, λ_2 and λ_3, its characteristic polynomial can be written as $(\lambda_1 - \lambda)(\lambda_2 - \lambda)(\lambda_3 - \lambda)$. Compare with the expression in (a)(i) to show that $a + e + i = \lambda_1 + \lambda_2 + \lambda_3$ and $\det(\boldsymbol{A}) = \lambda_1\lambda_2\lambda_3$.